

COMS 4720

1a) Number of possible games = $9! = 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 362880$

b)

Case I: X is in center

i) O is in a corner

a) $\begin{array}{ccc} O & - & - \\ - & X & - \\ - & - & - \end{array}$ b) $\begin{array}{ccc} - & - & O \\ - & X & - \\ - & - & - \end{array}$ c) $\begin{array}{ccc} - & - & - \\ - & X & - \\ O & - & - \end{array}$ d) $\begin{array}{ccc} - & - & - \\ - & X & - \\ - & - & O \end{array}$

ii) O is on a side

a) $\begin{array}{ccc} - & O & - \\ - & X & - \\ - & - & - \end{array}$ b) $\begin{array}{ccc} - & - & - \\ O & X & - \\ - & - & - \end{array}$ c) $\begin{array}{ccc} - & - & - \\ - & X & - \\ - & O & - \end{array}$ d) $\begin{array}{ccc} - & - & - \\ - & X & O \\ - & - & - \end{array}$

Case II: X in a corner

i) O is in a center

a) $\begin{array}{ccc} X & - & - \\ - & O & - \\ - & - & - \end{array}$ b) $\begin{array}{ccc} - & - & X \\ - & O & - \\ - & - & - \end{array}$ c) $\begin{array}{ccc} - & - & - \\ - & O & - \\ - & - & X \end{array}$ d) $\begin{array}{ccc} - & - & - \\ - & O & - \\ X & - & - \end{array}$

ii) O is on a side

a) $\begin{array}{ccc} X & O & - \\ - & - & - \\ - & - & - \end{array}$ b) $\begin{array}{ccc} X & - & - \\ O & - & - \\ - & - & - \end{array}$ c) $\begin{array}{ccc} - & O & X \\ - & - & - \\ - & - & - \end{array}$ d) $\begin{array}{ccc} O & - & - \\ - & - & - \\ - & - & - \end{array}$

iii) O in opp. corner

a) X - - b) - - X c) O - - d) - - O
 - - - - -
 - - O O - - - - X X - -

(Case 3: X is in a side

i) O is in the center

a) - X - b) - - - c) - - - d) - - -
 - O - - O - X O - - O X
 - - - - X - - - - -

ii) O is in a adjacent corner

a) O X - b) - X O c) - - - d) - - -
 - - - - -
 - - - - - O X - - - X O

iii) O is in the opp. side

a) - X - b) - O - c) - - - d) - - -
 - - - - - X - O O - X
 - O - - X - - - - -

c) In part b we there are 4 ways in which each can be addressed, but for this part let us focus on part a of each case to make things easier.

Case	Number of Os	Number of Xs	Evaluation
X is in center and O is in corner	There is 1 O in row 1 and col 1 and there is X in row 2 and col 2 ($O_1 = 2$)	There is X in row 2 and col 2. Plus there is one more X in the diagonal ($X_1 = 3$)	$3(0) + 3 - (0 + 2) = 1$
X is in center and O is on a side	There is 1 O in row 1 and in col 2 it gets blocked by X	There is one X in row 2 and 1 X appears in both the diagonals. ($X_1 = 3$)	$3(0) + 3 - (0+1) = 2$
X is in a corner and O is in the center	There is 1 O in col 2 and row 2 and it also appears in one of the diagonals (4)	There is one X in row 1 and col 1 (2).	$3(0) + 2 - (0+3) = -1$
X is in a corner and O is in the side	There is 1 O in col 2 (1)	There is one X in col 1 and it also appears in one of the diagonals (2).	$3(0) + 2 - (0+1) = 1$
X is in a corner and O is in the opposite corner	There is 1 O in each row 3 and col 3 (2)	There is one X in row 1 and col 1 (2)	$3(0) + 2 - (0+2) = 0$
X is in a side and O is in the adjacent corner	There is 1 O in col 1 and in the one of the diagonals (2)	There is 1 X in col 2 (1)	$3(0) + 1 - (0+2) = -1$
X is in a side and O is in the center	There is 1 O in row 2 and it also appears in both the diagonals (3)	There is 1 X in row 1 (1)	$3(0) + 1 - (0+3) = 1-3 = -2$
X is in side and O is in the opposite side	There is 1 O in row 2 (1)	There is 1 X in row 1	$3(0) + 1 - (0+1) = 0$

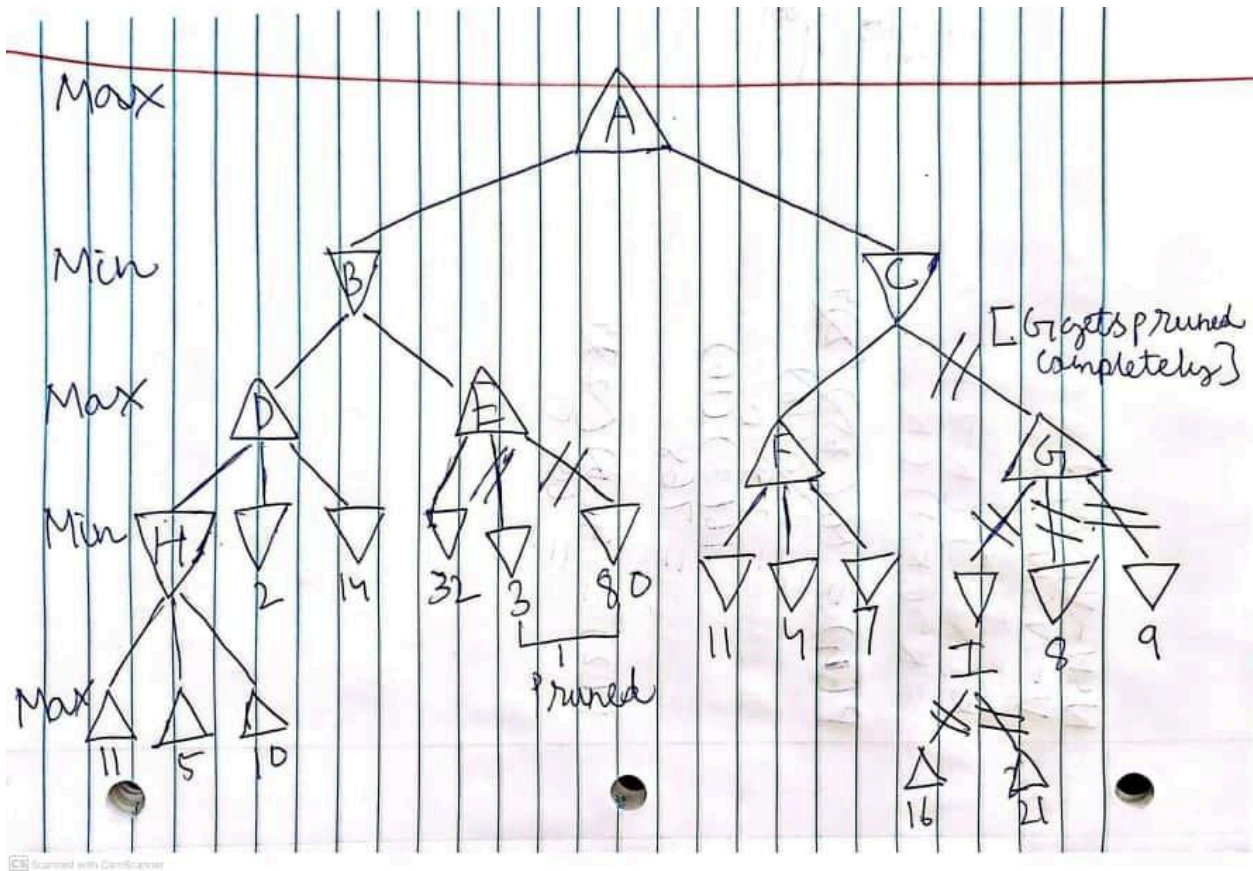
d) Let us assume that X plays at depth 0 and O replies at depth 1.

X's move	O's reply	Depth 1 value	Depth 0 value
Center	O can reply back by playing at a corner or on a side [Corner = +1, side = +2]	min (+1,+2)	+1
Corner	O can reply back by playing at a center or on a side or in opp. corner [Center = -1, side = +1, opp. corner = 0]	min(-1,+1,0)	-1
Side	O can reply back by playing at a center or on the adjacent corner or in opp. side [Center = -2, adj corner = -1, opp. side = 0]	min (-2,-1,0)	-2

In order to determine the best move for X, we must take out the maximum of all the depth 0 values: $\max(1,-1,-2) = +1$

Hence, we can say that the best move that X can play in the beginning is to move to the center.

2a)



2b)

Node	Max/Min	Children	Answer	(α, β) at return
H	MIN	11,5,10	5	$[-\infty, 5]$
D	MAX	5,2,14	14	$[14, +\infty]$
E	MAX	32,3,80	32 (we just look at the first child, since this operation returns the maximum which is definitely greater than 14, so it doesn't matter)	$[32, 14]$
B	MIN	14,32	14	$[-\infty, 14]$
F	MAX	11,4,7	11	$[14, +\infty]$
C	MIN	Since I and G get pruned out as both the children of I (16 and 21) are greater than 11 and C requires min value unlike G which requires a max value, so we get our answer here itself, which is 11.	11	$[14, 14]$
A	MAX	14,11	14	$[14, +\infty]$

c) The Max value at the root is 14.

3) First, let us take note of the regions and their neighbors as shown in the map.

Region	Neighbors
South Australia (SA)	Western Australia, Northern Territory, Queensland, Victoria, and New South Wales
Western Australia (WA)	Northern Territory and South Australia
Northern Territory (NT)	Western Australia, South Australia, and Queensland
Queensland (Q)	Northern Territory, New South Wales, and South Australia
New South Wales (NSW)	Victoria, South Australia, and Queensland
Victoria (V)	New South Wales, South Australia
Tasmania (T)	None

Let us consider that we have k colors, in which we can show this mapping. This means that SA and T can be given color since they share their borders with each one of them, or with none of them, which means that they have k choices to make. Next, if we move onto to NT, we know that it shares borders with SA, it can only have $k-1$ choices of colors since we don't want any neighboring to be of the same color. Similarly, we can say that WA, Q, NSW and V will have $(k-2)$ choices since they share borders with SA and one other state in common.

Let us consider a function $f(k)$ which can be determined as follows:

$$f(k) = k * (k - 1) * (k - 2) * (k - 2) * (k - 2) * (k - 2) * k \text{ (product of all the choices)}$$

$$f(k) = k^2 * (k - 1) * (k - 2)^4$$

Hence, we can say that there are $k^2 * (k - 1) * (k - 2)^4$ solutions for this problem.

Now, if $k=4$, we replace k with 4 in the function and do the calculation in the following way:

$$f(4) = 16 * (3) * 2^4$$

$$= 48 * 16 = 768$$

Therefore, there are 768 mapping solutions for this problem if there are 4 colors available to us.

Now, if $k=2$, we can replace k with 2 and since we know that $2-2=0$, the answer is going to be 0, which that it is not possible to solve this mapping problem with only 2 color choices.